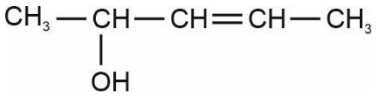
CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. How many different stereoisomers are possible for the given molecule?



(1) 2 (2) 4

(3) 3 (4) 1

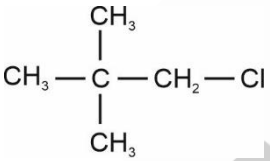
Answer (2)

Sol. n = 2

Number of stereoisomers = $(2)^2 = 4$

52. Given below are two statements:

Statement I : $CH_3 - O - CH_2 - Cl$ will undergo S_N1 reaction though it is a primary halide.

Statement II :  will not undergo S_N2 reaction very easily though it is a primary halide.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **Statement I** and **Statement II** are incorrect

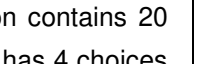
(2) Both **Statement I** and **Statement II** are correct

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

Answer (2)

Sol. Both statement-I and statement-II are correct

Statement – I :  is highly stable carbocation.

Statement – II : Due to hindrance, S_N2 will not take place easily.

53. A liquid when kept inside a thermally insulated closed vessel at 25ºC was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters?

(1) $\Delta U > 0, q = 0, w > 0$ (2) $\Delta U < 0, q = 0, w > 0$

(3) $\Delta U = 0, q < 0, w > 0$ (4) $\Delta U = 0, q = 0, w = 0$

Answer (1)

Sol. $q = 0$

$w > 0$

$\Delta U > 0$

54. In which of the following complexes the CFSE, Δ_0 will be equal to zero?

(1) $K_3[Fe(SCN)_6]$ (2) $[Fe(en)_3]Cl_3$

(3) $[Fe(NH_3)_6]Br_2$ (4) $K_4[Fe(CN)_6]$

Answer (1)

Sol. $K_3[Fe(SCN)_6]$ will have CFSE equal to zero (Fe^{3+} : $4s^0 3d^5 \left(t_{2g}^3 e_g^2\right)$

55. Which of the following electronegativity order is **incorrect**?

(1) Mg < Be < B < N (2) Al < Si < C < N

(3) Al < Mg < B < N (4) S < Cl < O < F

Answer (3)

Sol. Electronegativity of Al is higher than that of Mg

$\Rightarrow$ option (3) is correct



56. The **incorrect** statements regarding geometrical isomerism are

(A) Propene shows geometrical isomerism.

(B) Trans isomer has identical atoms/groups on the opposite sides of the double bond.

(C) Cis-but-2-ene has higher dipole moment than trans-but-2-ene.

(D) 2-methylbut-2-ene shows two geometrical isomers.

(E) Trans-isomer has lower melting point than cis isomer.

Choose the **correct** answer from the options given below.

(1) (A), (D) and (E) only (2) (A) and (E) only

(3) (C), (D) and (E) only (4) (B) and (C) only

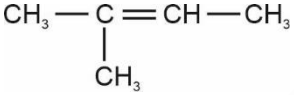
Answer (1)

Sol. Statement (A), (D) and (E) are incorrect statements.

(A) Propene does not show geometrical isomerism. (A) is incorrect.

(B) (B) is correct.

(C) Dipole moment : cis-but-2-ene > trans-but-2-ene

(D) 

2-methylbut-2-ene does not show geometrical isomerism. (D) is incorrect.

(E) Trans-isomer has higher melting point than cis-isomer. (E) is incorrect.

$\Rightarrow$ (A), (D) and (E) are incorrect.

57. From the magnetic behaviour of $[NiCl_4]^{2-}$ (paramagnetic) and $[Ni(CO)_4]$ (diamagnetic), choose the correct geometry and oxidation state.

(1) $[NiCl_4]^{2-}$: Ni^{II}, square planar

$[Ni(CO)_4]$: Ni(0), square planar

(2) $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral

$[Ni(CO)_4]$: Ni(0), tetrahedral

(3) $[NiCl_4]^{2-}$: Ni(0), tetrahedral

$[Ni(CO)_4]$: Ni(0), square planar

(4) $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral

$[Ni(CO)_4]$: Ni^{II}, square planar

Answer (2)

Sol. $[NiCl_4]^{2-} \Rightarrow Ni^{2+} : 4s^0 3d^8$

$\Rightarrow$ No pairing

$\Rightarrow sp^3$

$\Rightarrow$ Tetrahedral

$[Ni(CO)_4] \Rightarrow Ni^0 : 4s^0 3d^{10}$

$\Rightarrow sp^3$

$\Rightarrow$ Tetrahedral

58. Given below are two statements :

Statement I : One mole of propyne reacts with excess of sodium to liberate half a mole of H_2 gas.

Statement II : Four g of propyne reacts with $NaNH_2$ to liberate NH_3 gas which occupies 224 mL at STP.

In the light of the above statements, choose the **most appropriate answer** from the options given below.

(1) Both **Statement I** and **Statement II** are correct

(2) **Statement I** is incorrect but **Statement II** is correct

(3) **Statement I** is correct but **Statement II** is incorrect

(4) Both **Statement I** and **Statement II** are incorrect

Answer (3)

Sol. $CH_3 - C \equiv CH + Na \rightarrow CH_3 - C \equiv C^-Na^+ + \frac{1}{2}H_2$

Statement-I is correct.

Moles of $C_3H_4 = \frac{4}{40} = 0.1$ mole

$\underset{0.1\,\text{mole}}{CH_3 - C \equiv CH} + NaNH_2 \rightarrow CH_3 - C \equiv C^-Na^+ + \underset{0.1\,\text{mole}}{NH_3}$

Volume of $NH_3 = (0.1)(22.4) = 2.24$ L

Statement-II is incorrect.



59. Lanthanoid ions with $4f^7$ configuration are

(A) Eu^{2+} (B) Gd^{3+}

(C) Eu^{3+} (D) Tb^{3+}

(E) Sm^{2+}

Choose the correct answer from the options given below.

(1) (A) and (B) only

(2) (A) and (D) only

(3) (B) and (E) only

(4) (B) and (C) only

Answer (1)

Sol. Eu^{2+} and Gd^{3+} have $4f^7$ configuration.

60. Match List-I with List-II.

	List-I		List-II
(A)	$Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$	(I)	Ionisation enthalpy
(B)	B < C < O < N	(II)	Metallic character
(C)	B < Al < Mg < K	(III)	Electronegativity
(D)	Si < P < S < Cl	(IV)	Ionic radii

Choose the correct **answer** from the options given below.

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(2) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(4) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Answer (4)

Sol. (A) is order of ionic radii. A-(IV)

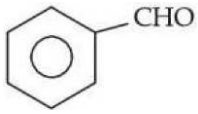
(B) is correct order of ionisation enthalpy. B-(I)

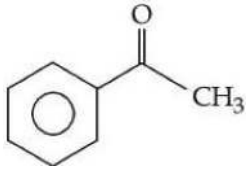
(C) is correct order of metallic character. C-(II)

(D) is correct order of electronegativity. D-(III)

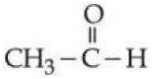
Option (4) is correct.

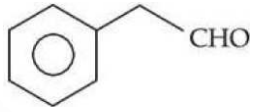
61. The compounds which give positive Fehling's test are:

(A) 

(B) 

(C) $HOCH_2 - CO - (CHOH)_3 - CH_2 - OH$

(D) 

(E) 

Choose the **correct** answer from the options given below:

(1) (C), (D) and (E) Only

(2) (A), (B) and (C) Only

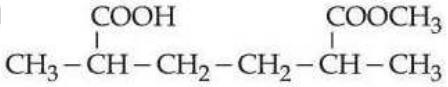
(3) (A), (D) and (E) Only

(4) (A), (C) and (D) Only

Answer (1)

Sol. Compound (C), (D) and (E) will give positive Fehling's test.

62. The IUPAC name of the following compound is:



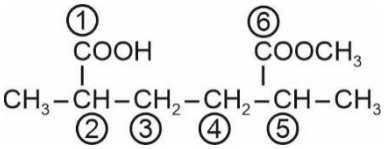
(1) Methyl-5-carboxy-2-methylhexanoate.

(2) 6-Methoxycarbonyl-2,5-dimethylhexanoic acid.

(3) Methyl-6-carboxy-2,5-dimethylhexanoate.

(4) 2-Carboxy-5-methoxycarbonylhexane.

Answer (2)

Sol. 

6-Methoxycarbonyl-2,5-dimethylhexanoic acid.



63. A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2A. The amount of the aluminium deposited at the cathode is _____.

[Given: molar mass of aluminium and chlorine are 27 g mol^{-1} and 35.5 g mol^{-1} respectively. Faraday constant = 96500 C mol^{-1}]

(1) 0.336 g (2) 0.441 g

(3) 1.007 g (4) 1.660 g

Answer (1)

Sol. $Al^{3+} + 3e^- \longrightarrow Al$

$$\text{Moles of electron} = \frac{2 \times 30 \times 60}{96500}$$

$$= \frac{36}{965}$$

$$\text{Moles of Al} = \frac{36}{3 \times 965}$$

$$= \frac{12}{965}$$

$$\text{Mass of Al} = \frac{12}{965} \times 27$$

$$= 0.336 \text{ gm}$$

64. Which of the following statement is not true for radioactive decay?

(1) Half life is ln 2 times of $\frac{1}{\text{rate constant}}$.

(2) Decay constant increases with increase in temperature.

(3) Amount of radioactive substance remained after three half lives is $\frac{1}{8}$th of original amount.

(4) Decay constant does not depend upon temperature.

Answer (2)

Sol. For radioactive decay, decay constant does not depend upon temperature because for radioactive decay activation energy is zero.

65. Radius of the first excited state of Helium ion is given as:

$a_0 \rightarrow$ radius of first stationary state of hydrogen atom.

(1) $r = \frac{a_0}{2}$ (2) $r = \frac{a_0}{4}$

(3) $r = 2a_0$ (4) $r = 4a_0$

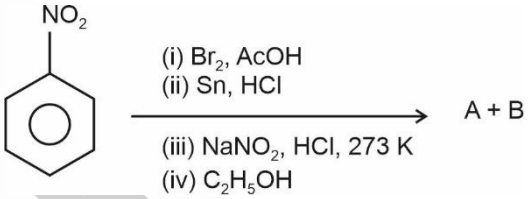
Answer (3)

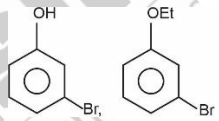
Sol. $r = a_0\left(\frac{n^2}{z}\right)$

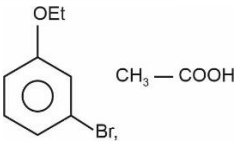
$$r = \frac{a_0(2)^2}{(2)}$$

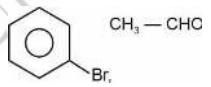
$r = 2a_0$

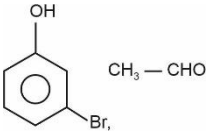
66. The products formed in the following reaction sequence are:



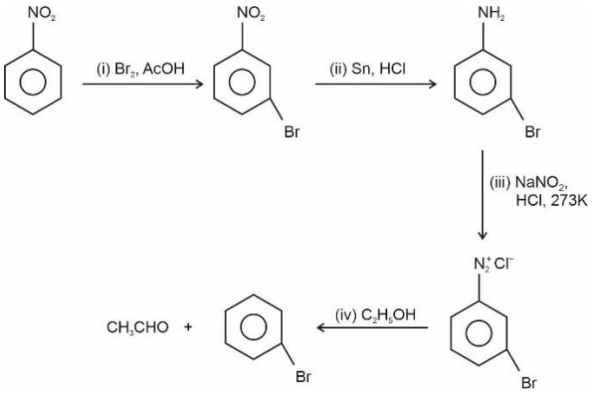
(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. 

Option (3) is correct



67. Which of the following electrolyte can be used to obtain $H_2S_2O_8$ by the process of electrolysis?

(1) Dilute solution of sulphuric acid

(2) Dilute solution of sodium sulphate

(3) Acidified dilute solution of sodium sulphate

(4) Concentrated solution of sulphuric acid

Answer (4)

Sol. $H_2S_2O_8$ is obtained by electrolysis of concentrated solution of sulphuric acid.

68. Arrange the following solutions in order of their increasing boiling points.

(i) 10^{-4} M NaCl (ii) 10^{-4} M Urea

(iii) 10^{-3} M NaCl (iv) 10^{-2} M NaCl

(1) (ii) < (i) ≡ (iii) < (iv) (2) (i) < (ii) < (iii) < (iv)

(3) (iv) < (iii) < (i) < (ii) (4) (ii) < (i) < (iii) < (iv)

Answer (4)

Sol. $\Delta T_b = i(K_b(m))$

ΔT_b order or T_b order

(iv) > (iii) > (i) > (ii)

69. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of CO_2 is converted into CO on addition of graphite. If total pressure at equilibrium is 0.8 atm, then Kp is:

(1) 0.18 atm (2) 1.8 atm

(3) 0.3 atm (4) 3 atm

Answer (2)

Sol.

$$CO_{2(g)} + C(s) \rightleftharpoons 2CO_{(g)}$$

0.5 atm	—
↓	↓
(0.5 – x)	(2x)

$(0.5 - x) + (2x) = 0.8$

$0.5 + x = 0.8$

$\boxed{x = 0.3}$

$$K_p = \frac{(0.6)(0.6)}{(0.2)} = 1.8 \text{ atm}$$

70. Which of the following acids is a vitamin?

(1) Saccharic acid (2) Adipic acid

(3) Aspartic acid (4) Ascorbic acid

Answer (4)

Sol. Vitamin – C is also known as Ascorbic Acid.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. In Carius method for estimation of halogens, 180 mg of an organic compound produced 143.5 mg of AgCl. The percentage composition of chlorine in the compound is _____%.

(Given : molar mass in g mol^{-1} of Ag : 108, Cl : 35.5)

Answer (20)

Sol. millimoles of AgCl = $\frac{143.5}{143.5} = 1$

Millimoles of Cl = 1

Mass of Cl = 35.5×10^{-3} gm

% by mass of Cl = $\frac{35.5}{180} \times 100$

= 19.72%

Nearest integer = 20%

72. Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed through an aqueous solution of $Ca(OH)_2$. The excess unreacted $Ca(OH)_2$ was later neutralized with 0.1 M of 40 mL HCl. If the volume of the sealed container of CO_2 was x, then x is _____cm^3 (nearest integer).

[Given : The entire amount of $CO_2(g)$ reacted with exactly half the initial amount of $Ca(OH)_2$ present in the aqueous solution]



Answer (45)

Sol. moles of HCl = 4×10^{-3}

moles of $Ca(OH)_2$ reacted with HCl = 2×10^{-3}

Initial amount of $Ca(OH)_2$ = 4×10^{-3} moles

Moles of CO_2 = 2×10^{-3}

Volume = $2 \times 10^{-3} \times 22.4$

= 44.8×10^{-3} Lit

= 44.8 cm^3

Nearest integer = 45

73. The number of molecules/ions that show linear geometry among the following is _____.

$SO_2, BeCl_2, CO_2, N_3^-, NO_2, F_2O, XeF_2, NO_2^+, I_3^-, O_3$

Answer (6)

Sol. linear geometry is shown by

$BeCl_2; CO_2; N_3^-; XeF_2; NO_2^+; I_3^-$

74. $A \rightarrow B$

The molecule A changes into its isomeric form B by following a first order kinetics at a temperature of 1000 K. If the energy barrier with respect to reactant energy for such isomeric transformation is 191.48 kJ mol^{-1} and the frequency factor is 10^{20}, the time required for 50% molecules of A to become B is ___ picoseconds (nearest integer). [R = 8.314 J K^{-1} mol^{-1}]

Answer (69)

Sol. $k = Ae^{-\frac{Ea}{RT}}$

$= (10^{20})\left(e^{-\frac{191.48\times1000}{(8.314)(1000)}}\right)$

$= 10^{20}\ (e^{-23.03})$

$= (10^{20})(10^{-10})$

$= 10^{10}$

$t_{1/2} = \frac{0.693}{10^{10}}$

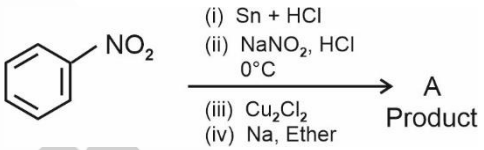
$= 0.693 \times 10^{-10}$

$= 69.3 \times 10^{-12}$

= 69.3 picoseconds

Nearest integer = 69

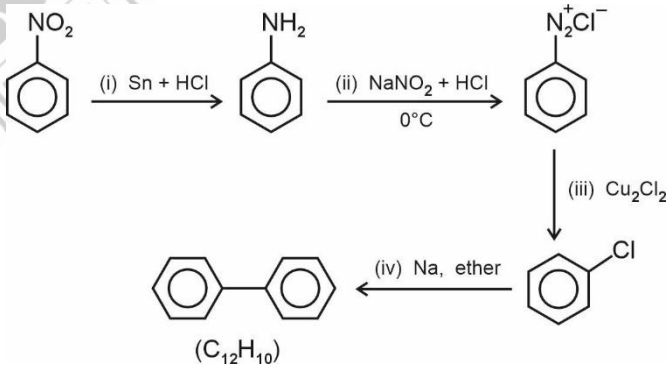
75. Consider the following sequence of reactions :



Molar mass of the product formed (A) is _____ g mol^{-1}.

Answer (154)

Sol.



Molar mass of A = $154 \frac{gm}{mole}$

❑ ❑ ❑

